

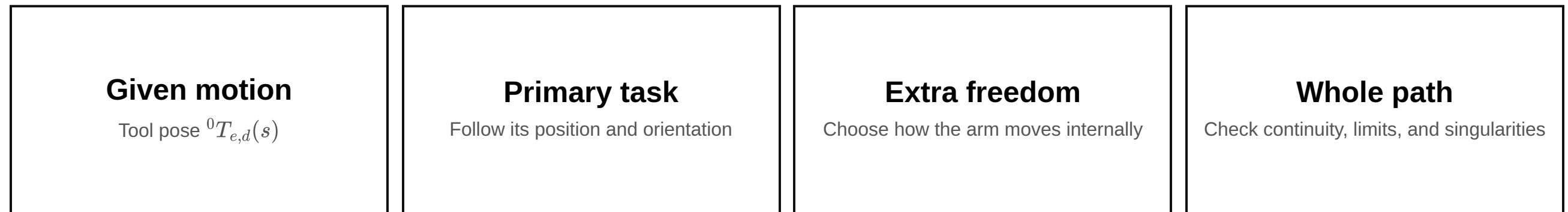
Redundancy Resolution in Motion Planning

KUKA iiwa 7 · one supplied workspace motion, many joint motions

Understand the null space, compare cost functions, and explore numerical and analytical strategies on the same rectangular path.

The workspace command is given; we choose the joint motion

As in Lecture 07, a precomputed trajectory or a real-time policy supplies the desired motion. Here we study how redundancy changes the planning problem.



Our example is a rectangular path with fixed tool orientation. Path progress $s \in [0, 1]$ identifies where we are on the supplied command.

At a regular pose, the iiwa has one instantaneous internal freedom

$${}^0\mathcal{V}_e = \begin{bmatrix} {}^0\dot{\mathbf{p}}_e \\ {}^0\boldsymbol{\omega}_e \end{bmatrix} = {}^0J_e(\mathbf{q})\dot{\mathbf{q}}$$

${}^0J_e \in \mathbb{R}^{6 \times 7}$, with linear velocity first.

Rank counts independent task velocities

If $\text{rank } J = 6$, every infinitesimal tool velocity can be produced locally.

Nullity counts internal velocity directions

$$\dim \ker J = 7 - \text{rank } J = 1$$

A nonzero combination of joint rates can leave the tool velocity equal to zero.

There is a continuous family of IKs near a regular solution. Joint limits and singularities can restrict this family and its connections along the path.

Every joint contributes a screw evaluated at the tool point

1 · Locate the axis

Multiply the URDF transforms along the chain to obtain a point ${}^0\mathbf{p}_i$ on joint axis i and the joint-frame rotation ${}^0R_{J_i}$.

2 · Express it in the world frame

$${}^0\mathbf{a}_i = {}^0R_{J_i} J_i \hat{\mathbf{a}}_i$$

Transform the axis written in the URDF into the same frame as the tool point.

3 · Take the cross product

$${}^0J_{e,i} = \begin{bmatrix} {}^0\mathbf{a}_i \times ({}^0\mathbf{p}_e - {}^0\mathbf{p}_i) \\ {}^0\mathbf{a}_i \end{bmatrix}$$

Stack all seven columns in joint order.

$${}^0J_e = \begin{bmatrix} {}^0J_{e,1} & {}^0J_{e,2} & \cdots & {}^0J_{e,7} \end{bmatrix}$$

We use angular velocity, not Euler-angle rates. The robot models and the analytical IK are checked against the same native KUKA iiwa 7 geometry.

Different joint-rate vectors can create the same tool velocity

$$J\dot{\mathbf{q}}_a = \mathcal{V}_d, \quad J\dot{\mathbf{q}}_b = \mathcal{V}_d$$

$$J(\dot{\mathbf{q}}_b - \dot{\mathbf{q}}_a) = \mathbf{0}$$

Subtract the two solutions

Their difference is an internal motion: its joint contributions cancel at the tool.

Add any such motion

$$\dot{\mathbf{q}} = \dot{\mathbf{q}}_p + \mathbf{n}\nu, \quad J\mathbf{n} = \mathbf{0}$$

At full row rank, one scalar ν selects the velocity along the iiwa's local self-motion direction.

A null-space direction usually moves several joints together. It is not a single joint that can be moved independently while keeping the tool fixed.

The pseudoinverse follows the task; the projector selects internal motion

One particular solution

$$J^+ = J^T(JJ^T)^{-1}$$

For rank six, $J^+\mathcal{V}_d$ is the solution with minimum Euclidean joint-rate norm.

Project into the null space

$$N = I - J^+J$$

$JN = 0$, $N^2 = N$, and $N^T = N$. For one redundant degree, $N = \mathbf{n}\mathbf{n}^T$ if $\|\mathbf{n}\| = 1$.

All instantaneous solutions

$$\dot{\mathbf{q}} = J^+\mathcal{V}_d + N\mathbf{z}$$

The free vector $\mathbf{z}$ expresses our secondary preference.

These identities describe the exact pseudoinverse. Damped least squares improves numerical conditioning, but its associated matrix is generally not an exact null-space projector.

Change the redundant angle and watch the arm reconfigure

Hold the tool at the rectangle's centre, ${}^0\mathbf{p}_e = (0.45, 0, 0.60)$ m, with its orientation fixed. This pose has eight continuous IK branches over $q_3 = -170^\circ$ to $+170^\circ$. Follow the other six joint angles; grey portions exceed joint limits.

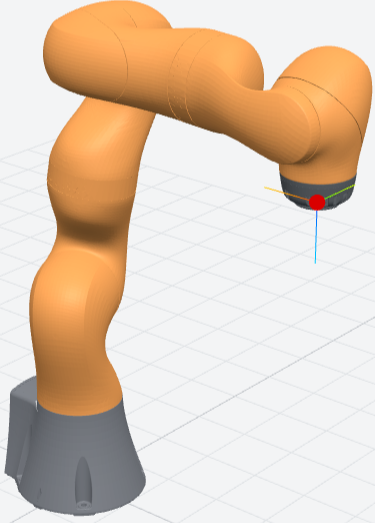
Fixed tool pose · centre

x0.45y0.00z0.60m

Rebuild curvesDefault

KUKA iiwa 7 · native joint limits

tool frame held fixed



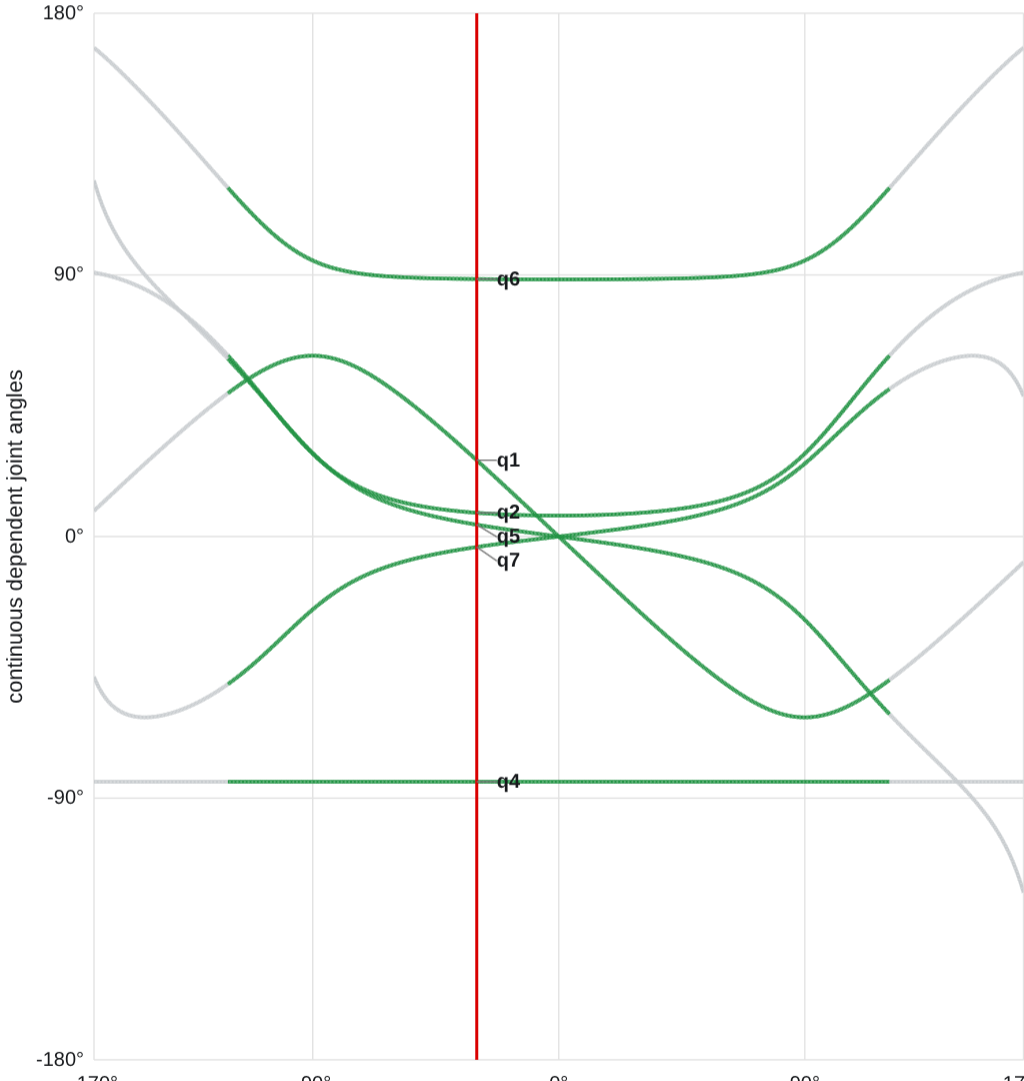
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Continuous self-motion · fixed pose

☐ All 8 IK curves

IK 5 · continuous joint angles, fixed tool pose



continuous dependent joint angles

redundant angle q_3

● within joint limits

● outside joint limits

● selected q_3

Redundant angle

-30°

IK 1 joint limit

IK 2 joint limit

IK 3 joint limit

IK 4 joint limit

IK 5 within limits

IK 6 joint limit

IK 7 joint limit

IK 8 joint limit

$\mathbf{J} \dot{\mathbf{q}} = 0$

7 joint velocities · 6 tool constraints

IK 5 · $q_3 = -30.0^\circ$

tool position error $1.2\text{e-}16$ m

tool rotation error $9.0\text{e-}16$ rad

minimum joint margin 31.4°

Full J: regular · nullity 1

$\mathbf{q} = 26, 8, -30, -84, 4, 89, -4^\circ$

Move q_3 along a continuous fixed-pose curve. Grey sections are mathematical configurations beyond the robot's joint limits.

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Different objectives choose different internal motions

Let $q_{i,c} = (q_{i,\max} + q_{i,\min})/2$, $r_i = (q_{i,\max} - q_{i,\min})/2$, and $u_i = (q_i - q_{i,c})/r_i$. The reference posture is the selected starting IK.

Interpret u_i : it is the joint's dimensionless position relative to the centre of its allowed range. $u_i = 0$ is the centre, $u_i = -1$ is the lower limit, and $u_i = 1$ is the upper limit. Thus $|u_i| > 1$ means the joint is outside its limits; this normalization lets us compare joints with different ranges.

Dropdown strategy	Cost to minimize	What the arm prefers
No secondary cost	$\mathbf{z} = \mathbf{0}$	Minimum joint-rate norm at each step; no explicit joint-limit avoidance.
Joint centering	$h_c = \frac{1}{2} \sum_i u_i^2$	Stay near the centres of the joint ranges.
Joint-limit barrier	$h_b = -\sum_i \log(1 - u_i^2)$	Increasingly penalize approach to a physical stop, for $ u_i < 1$.
Reference posture	$h_r = \frac{1}{2} \ \mathbf{q} - \mathbf{q}_{\text{ref}}\ ^2$	Remain close to the selected starting posture.

The secondary cost acts only through the available null direction

$$\dot{\mathbf{q}} = J^+ \mathcal{V}_d - \alpha N \nabla h(\mathbf{q}), \quad \alpha \geq 0$$

Hold the task fixed

$$\mathcal{V}_d = 0 \quad \Rightarrow \quad \dot{h} = -\alpha \nabla h^T N \nabla h = -\alpha \|N \nabla h\|^2 \leq 0$$

The exact projected gradient decreases the cost unless its projection vanishes.

Move along the prescribed path

$$\dot{h} = \nabla h^T J^+ \mathcal{V}_d - \alpha \|N \nabla h\|^2$$

The primary task may increase the cost faster than the secondary motion can reduce it.

A cost function is a preference, not a guarantee. The null-space direction may offer no escape from a joint limit imposed by the primary task.

Predict with the Jacobian, correct the pose, then check the step

- 1 Select one legal starting IK from the map and choose the null-space objective and its gain.
- 2 Compute the required tool displacement and the current Jacobian. Add the projected secondary motion to the pseudoinverse prediction.
- 3 Correct the finite-step position and orientation error. A first-order null prediction alone drifts away from a curved self-motion manifold.
- 4 Validate joint limits, full-Jacobian rank, pose accuracy, and continuity along the step. Stop and report the first failed check.

$$\mathbf{q}_{k+1}^{\text{pred}} = \mathbf{q}_k + J_k^+ \Delta \mathbf{x}_k - \alpha \Delta s N_k \nabla h(\mathbf{q}_k)$$

Here $\Delta \mathbf{x}_k = [\Delta \mathbf{p}_k; \text{Log}(R_{k+1} R_k^T)^\vee]$ is a small geometric pose increment, with both components expressed in the world frame.

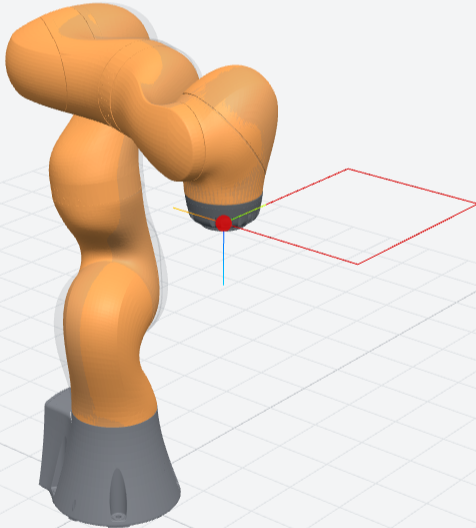
Follow the same rectangle with different starting IKs and costs

Select a starting point at $s = 0$, choose the null-space cost, and build the numerical path. The default rectangle includes a starting solution that completes the path within the joint limits.

Shared rectangle · fixed tool orientation

KUKA iiwa 7 · native joint limits

black: given path · red: followed path

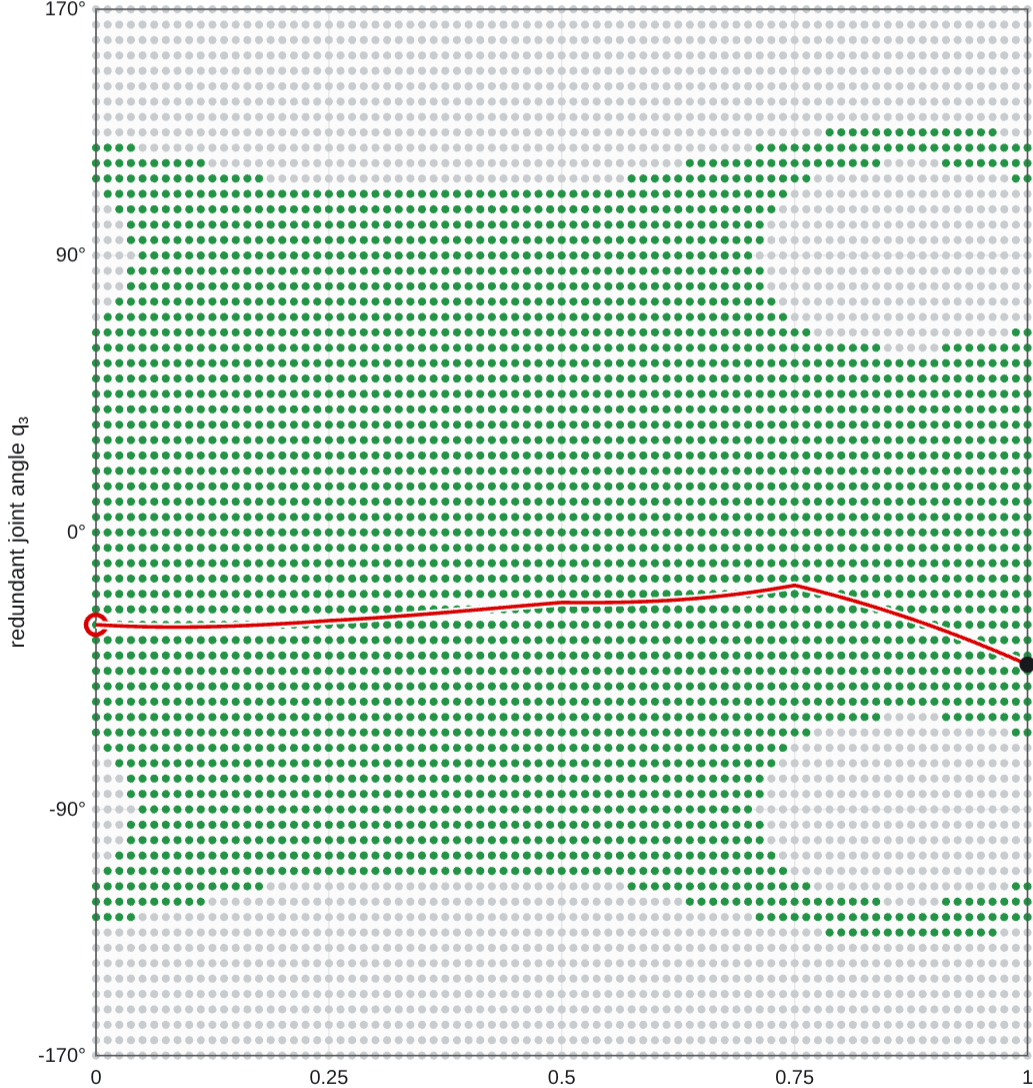


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Path progress × redundant angle

IK 5 · click a green start at $s = 0$



workspace path progress s

● legal, regular ● limit / singular / no root ● plan on this IK chart

Starting redundant angle

-30°

IK 1

limit / rank

IK 2

limit / rank

IK 3

limit / rank

IK 4

legal start

IK 5

legal start

IK 6

limit / rank

IK 7

limit / rank

IK 8

limit / rank

Planner

Numerical · null space

Null-space objective

Joint centering

Null-space gain

0.80

Build path

Compare 8 starts

Replay

Reset

IK 5 · numerical / centering

$s = 1.000$ · $q_3 = -42.9^\circ$

COMPLETE PATH

minimum limit margin 20.0°

max pose error $2.0e-10$

min $\sigma(\hat{J})$ 0.1417

Complete: this start and strategy follow all four edges within the checked limits and singularity margin.

Fixing the redundant angle turns a continuous family into discrete candidates

$$\mathcal{M}_s = \{\mathbf{q} : f(\mathbf{q}) = {}^0T_{e,d}(s)\}$$

$$\mathcal{S}_s(\lambda) = \mathcal{M}_s \cap \{q_3 = \lambda\}$$

Before fixing q_3

A reachable regular pose has a locally one-dimensional inverse image.

After fixing $q_3 = \lambda$

The iiwa model generically has up to eight isolated analytical IK roots. A particular pose and angle can have fewer roots, and some roots violate joint limits.

Solve the arm geometry, then the wrist orientation

1 · Two elbow choices

Let $\mathbf{w}$ run from the shoulder to the wrist centre. The two arm lengths are $a = b = 0.4$ m.

$$c_4 = \frac{\|\mathbf{w}\|^2 - a^2 - b^2}{2ab}$$

$$q_4 = \pm \arccos c_4$$

2 · Two shoulder choices

With $q_3 = \lambda$, set $A = a + bc_4$ and $B = -bs_4 \cos \lambda$.

$$w_z = A \cos q_2 - B \sin q_2$$

This gives two generic q_2 roots. Then use the horizontal wrist coordinates to recover q_1 .

3 · Two wrist choices

Remove the arm rotation from the desired tool rotation:

$$W = ({}^0R_4)^{T0} R_{e,d}$$

Resolve this spherical-wrist rotation into two generic (q_5, q_6, q_7) solutions.

$$2 \times 2 \times 2 = 8 \text{ generic fixed-angle branches}$$

The three $\pm$ signs label fixed-angle algebraic roots; they are not global SEW classes for the redundant robot. Each root is checked by forward kinematics and native joint limits before planning its continuation.

Every dot is an IK evaluation at one path position and redundant angle

Horizontal · path progress

$s \in [0, 1]$ moves around the rectangle. The same position and orientation command applies to every candidate.

Vertical · redundant angle

$\lambda = q_3$ selects a fixed-angle IK slice. Inspect each of the eight branches separately or show all branches.

Green and grey dots

Green: a real root passes the implemented pose, joint-limit, and full-rank checks. **Grey:** no such admissible root at that sample.

A row keeps q_3 constant. A curve through the map changes redundancy along the path. Green dots certify local samples; the connections between them must also be checked.

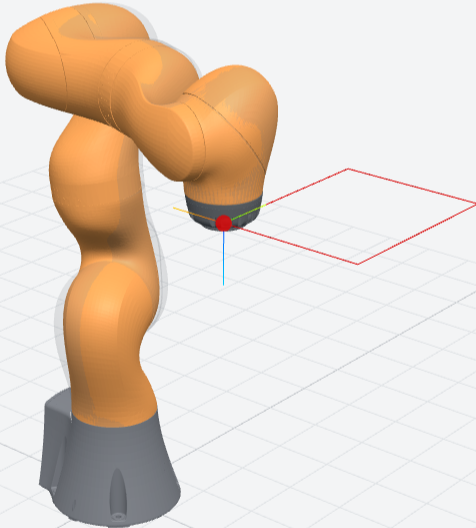
Click a starting IK on the map and compare its possible futures

Green and grey dots come from analytical IK evaluations. Select a legal dot at $s = 0$, then follow it numerically with your chosen null-space cost. Edit the rectangle and rebuild the map to explore a different command.

Shared rectangle · fixed tool orientation

KUKA iiwa 7 · native joint limits

black: given path · red: followed path

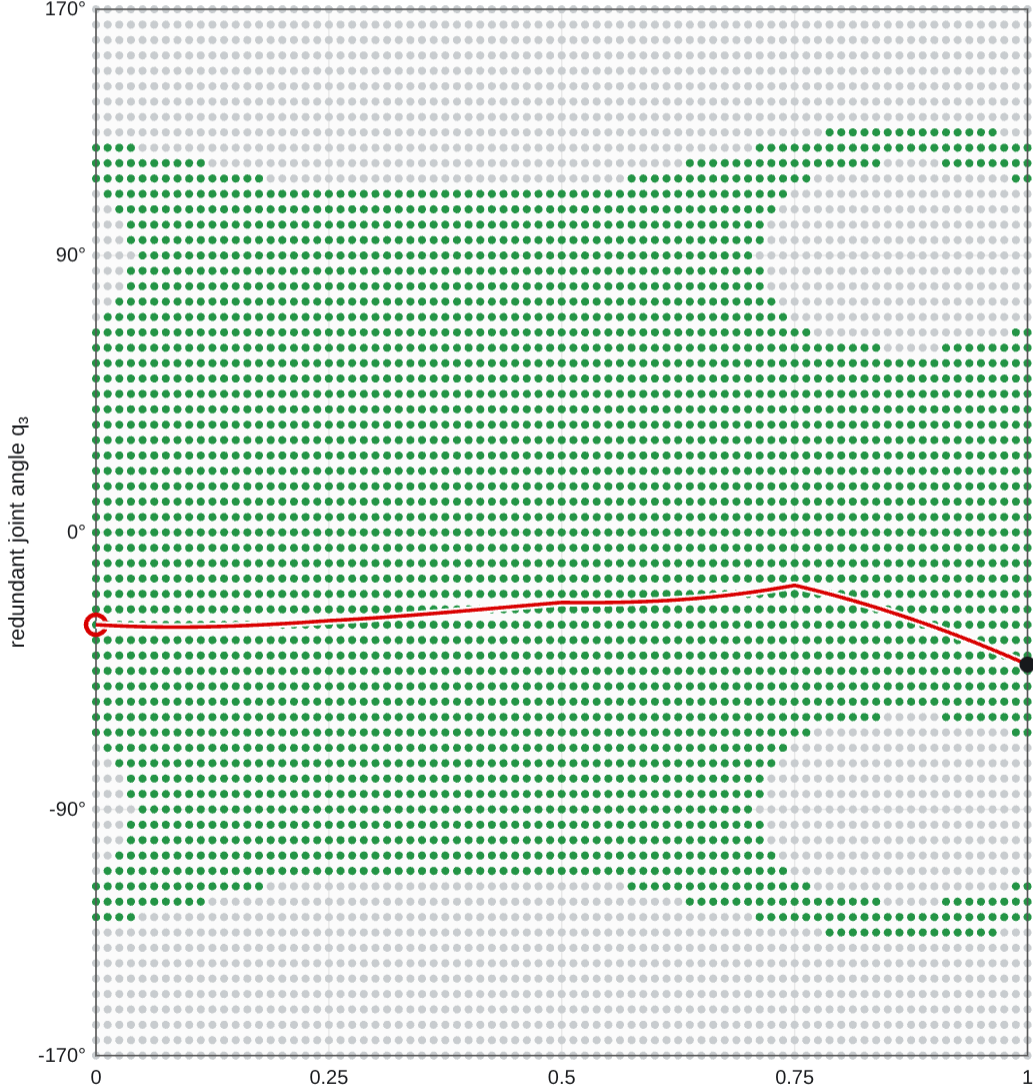


STL opacity

100%

Path progress × redundant angle

IK 5 · click a green start at $s = 0$



● legal, regular

● limit / singular / no root

● plan on this IK chart

Starting redundant angle

-30°

IK 1

limit / rank

IK 2

limit / rank

IK 3

limit / rank

IK 4

legal start

IK 5

legal start

IK 6

limit / rank

IK 7

limit / rank

IK 8

limit / rank

Planner

Numerical · null space

Null-space objective

Joint centering

Null-space gain

0.80

Build path

Compare 8 starts

Replay

Reset

IK 5 · numerical / centering

$s = 1.000 \cdot q_3 = -42.9^\circ$

COMPLETE PATH

minimum limit margin 20.0°

max pose error 2.0e-10

min $\sigma(\hat{J})$ 0.1417

Complete: this start and strategy follow all four edges within the checked limits and singularity margin.

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Search connected IK candidates across the whole rectangle

Nodes

$$\mathbf{q}_k^{(b,r)} \in \mathcal{S}_{s_k}(\lambda_r)$$

At each path layer, enumerate the analytical roots for every sampled redundancy angle.

Edges

Connect nearby configurations in adjacent layers only after checking continuous joint motion, pose tracking, limits, and singularity margin.

Proximity on the 2D plot alone is insufficient.

Whole-path strategy

Search from the selected starting IK to the final layer. Compare joint travel and clearance from joint limits across complete candidate paths.

Analytical planning evaluates the global candidate structure. Numerical continuation resolves motion locally. A failed search on a finite grid does not prove that no continuous path exists.

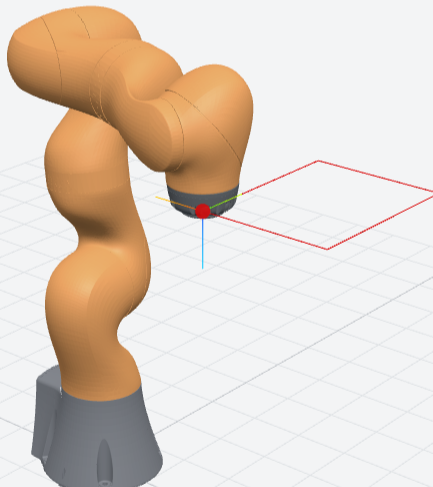
Choose a redundancy schedule using information from the entire path

With the default rectangle and $q_3 = -30^\circ$, try IK 4: numerical joint centering stops, but global planning finds a complete route. Compare the joint-travel and joint-limit objectives, then play their redundancy schedules.

Shared rectangle · fixed tool orientation

KUKA iiwa 7 · native joint limits

black: given path · red: followed path



STL opacity 100%

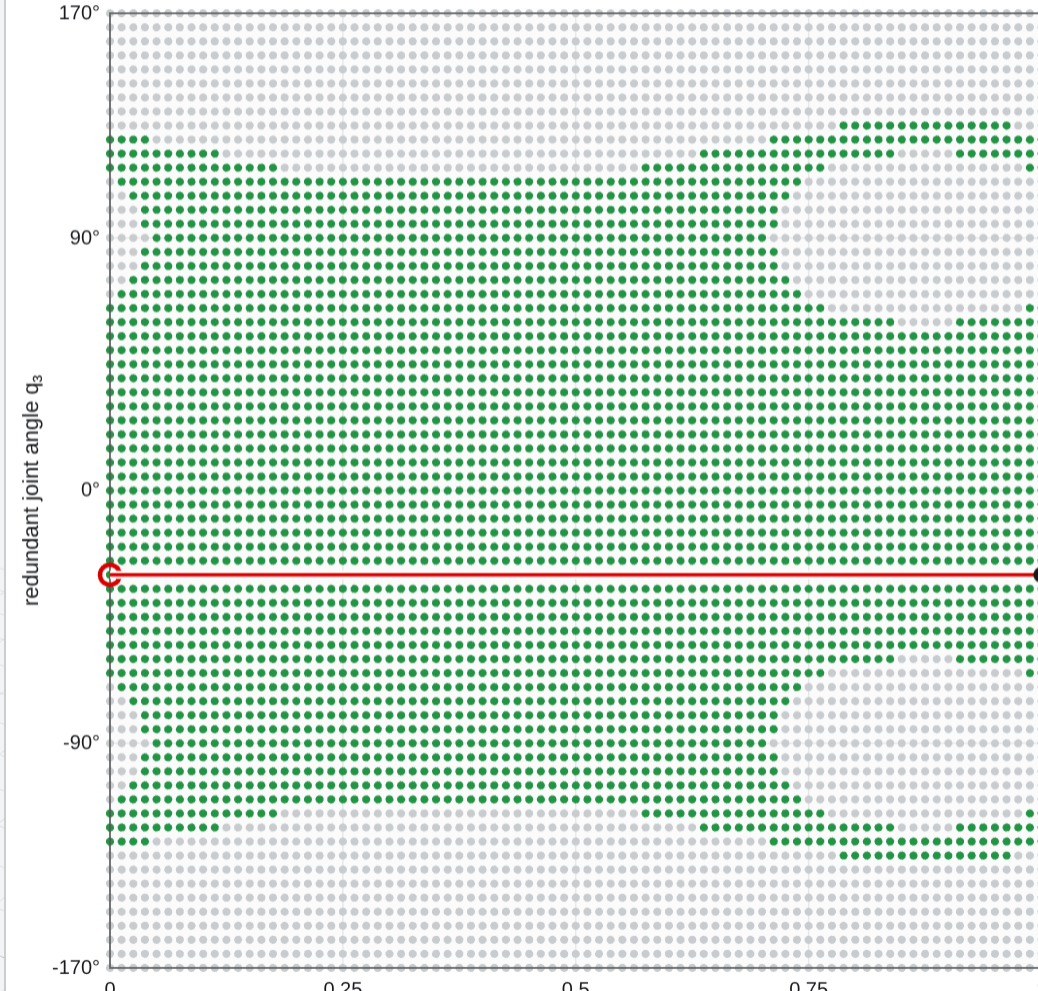
x 0.45y 0.00z 0.60width 0.22height 0.28m

Rebuild mapDefault

Path progress × redundant angle

☐ All 8 maps

IK 5 · click a green start at s = 0



● legal, regular

● limit / singular / no root

● plan on this IK chart

Starting redundant angle

-30°

IK 1

limit / rank

IK 2

limit / rank

IK 3

limit / rank

IK 4

legal start

IK 5

legal start

IK 6

limit / rank

IK 7

limit / rank

IK 8

limit / rank

Planner

Analytical · global map

Global edge cost

Least joint travel

Build path

Compare 8 starts

Replay

Reset

IK 5 · analytical / travel

s = 1.000 · q3 = -30.0°

COMPLETE PATH

minimum limit margin 20.0°

max pose error 3.6e-15

min σ(J) 0.1412

Complete: this start and strategy follow all four edges within the checked limits and singularity margin.

Check the full 6×7 Jacobian before declaring a lost task direction

Fixed-angle chart

$$\bar{J} = J_{[:, 1, 2, 4, 5, 6, 7]} \in \mathbb{R}^{6 \times 6}$$

$\det \bar{J} = 0$ means that holding q_3 fixed prevents local inversion of the remaining six joints. The robot may still have full task rank.

Full redundant robot

$$\text{rank } J < 6 \iff \sigma_{\min}(\hat{J}) = 0$$

The full robot then loses an independent instantaneous task velocity. The redundant coordinate may need to change before this boundary is reached.

Concrete iiwa example: $\mathbf{q} = (20, -35, 30, 50, -40, 0, 60)^\circ$ is within the joint limits. The full scaled Jacobian has $\sigma_{\min} = 0.0477$ m and rank six; removing the q_3 column gives rank five. Here q_3 fails as a local parameter while the tool retains all six instantaneous velocity directions.

Change one choice at a time and inspect the complete joint path

- 1** Keep the rectangle fixed. At one redundant angle, inspect all eight analytical roots and identify the legal starting IKs.
- 2** Keep the start fixed. Compare no secondary cost, joint centering, the joint-limit barrier, and reference-posture attraction.
- 3** Compare numerical continuation with an analytical graph search. Inspect how the chosen redundant angle changes along the path.
- 4** Move or enlarge the rectangle. Rebuild the maps and investigate which starts and strategies still complete the command.

For a closed workspace loop, compare the final and initial joint configurations. Following the same tool loop need not return a redundant arm to the same posture.

Redundancy resolution chooses a continuous route through the IK family

Understand the freedom

The null space identifies instantaneous internal motion that preserves the task velocity.

Choose a local preference

A projected cost gradient selects how numerical planning uses that internal freedom.

Check the global connections

The analytical path–angle maps expose alternative starts and redundancy schedules. Validate the entire connection before accepting a plan.

These demonstrations check kinematic tracking, native joint limits, and singularity margins. Collision checking, dynamics, and a physically smooth time law are additional requirements for execution on hardware.